## Contribution submission for Deutsche Physikerinnentagung 2016

# Towards the Development of Transparent Graphene Micro Electrode Arrays

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Microelectrode arrays (MEAs) are widely used to record the cellular activity. The standard electrode materials in use are platinum, gold and titanium nitride. The entry of carbon based materials in the field of MEAs is only a decade old. Since its discovery, graphene is one of the most extensively researched material. Application of graphene in optogenetics [1] and calcium imaging [2] is reported in the year 2014. However, the electrode size in both cases are significantly large. The main advantage of graphene is transparency, which comes at the cost of impedance. Here, we present the fabrication of graphene-based microelectrode arrays comprising gold interconnects terminated by graphene electrodes, where an optimal balance between transparency, size of the electrode and impedance is achieved.

Graphene is grown by chemical vapour deposition using copper as catalyst and methane as the carbon source. It is subsequently transferred on the conduction lines via polymer-based-transfer technique. Raman spectroscopy confirms the presence of monolayer over the 30 micron diameter electrode. In addition, the impedance spectrum shows ohmic contact between the gold conduction lines and graphene.

[1] D.-W. Park et. al., Nat. Comm., 5, 5258 (2014)

[2] D. Kuzum et. al., Nat. Comm., 5, 5259 (2014)

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